

✓ INF05731 Assignment: 4

This exercise will provide a valuable learning experience in working with text data and extracting features using various topic modeling algorithms. Key concepts such as Latent Dirichlet Allocation (LDA), Latent Semantic Analysis (LSA) and BERTopic.

Expectations:

- Students are expected to complete the exercise during lecture period to meet the active participation criteria of the course.
- Use the provided `.ipynb` document to write your code & respond to the questions. Avoid generating a new file.
- Write complete answers and run all the cells before submission.
- Make sure the submission is "clean"; *i.e.*, no unnecessary code cells.
- Once finished, allow shared rights from top right corner (*see Canvas for details*).

Total points: 100

NOTE: The output should be presented well to get **full points**

Late submissions will have a penalty of 10% of the marks for each day of late submission, and no requests will be answered. Manage your time accordingly.

✓ Question 1 (20 Points)

Dataset: 20 Newsgroups dataset

Dataset Link: https://scikit-learn.org/0.19/datasets/twenty_newsgroups.html

Consider Random 2000 rows only

Generate K=10 topics by using LDA and LSA, then calculate coherence score and determine the optimized K value by the coherence score. Further, summarize and visualize each topics in your own words.

```
import nltk
from nltk.corpus import stopwords
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from nltk.stem import WordNetLemmatizer
import re

nltk.download('stopwords')
nltk.download('wordnet')

stop_words = set(stopwords.words('english'))
```

```
lemmatizer = WordNetLemmatizer()
```

```
def preprocess(text):
    text = re.sub(r'\W+', ' ', text.lower())
    tokens = text.split()
    tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in stop_words and
    return " ".join(tokens)
```

```
df['cleaned'] = df['text'].apply(preprocess)
```

➡ [nltk_data] Downloading package stopwords to /root/nltk_data...
 [nltk_data] Unzipping corpora/stopwords.zip.
 [nltk_data] Downloading package wordnet to /root/nltk_data...

```
!pip uninstall -y gensim numpy
```

➡ Found existing installation: gensim 4.3.3
 Uninstalling gensim-4.3.3:
 Successfully uninstalled gensim-4.3.3
 Found existing installation: numpy 1.26.4
 Uninstalling numpy-1.26.4:
 Successfully uninstalled numpy-1.26.4

```
!pip install numpy==1.24.4
```

➡ Requirement already satisfied: numpy==1.24.4 in /usr/local/lib/python3.11/dist-packag

```
!pip install gensim
```

➡ Collecting gensim
 Using cached gensim-4.3.3-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
 Requirement already satisfied: numpy<2.0,>=1.18.5 in /usr/local/lib/python3.11/dist-p
 Requirement already satisfied: scipy<1.14.0,>=1.7.0 in /usr/local/lib/python3.11/dist
 Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.11/dist-pa
 Requirement already satisfied: wrapt in /usr/local/lib/python3.11/dist-packages (from
 Using cached gensim-4.3.3-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
 Installing collected packages: gensim
 Successfully installed gensim-4.3.3

```
from sklearn.datasets import fetch_20newsgroups
import random
import pandas as pd
import numpy as np
# Load full dataset
data = fetch_20newsgroups(subset='all', remove=('headers', 'footers', 'quotes'))

# Sample 2000 random posts
random.seed(42)
```

```

indices = random.sample(range(len(data.data)), 2000)
sampled_data = [data.data[i] for i in indices]
df = pd.DataFrame(sampled_data, columns=["text"])

import nltk
from nltk.corpus import stopwords
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from nltk.stem import WordNetLemmatizer
import re

nltk.download('stopwords')
nltk.download('wordnet')

stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()

def preprocess(text):
    text = re.sub(r'\W+', ' ', text.lower())
    tokens = text.split()
    tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in stop_words]
    return " ".join(tokens)

df['cleaned'] = df['text'].apply(preprocess)

➡ [nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!

from sklearn.decomposition import LatentDirichletAllocation, TruncatedSVD
from gensim.models.coherencemodel import CoherenceModel
from gensim.corpora.dictionary import Dictionary
import gensim
import numpy as np

# Tokenized docs
tokenized_docs = [doc.split() for doc in df['cleaned']]

# Create Dictionary and Corpus
dictionary = Dictionary(tokenized_docs)
corpus = [dictionary.doc2bow(text) for text in tokenized_docs]

# Create TF and TF-IDF matrices
vectorizer = CountVectorizer(max_df=0.95, min_df=2)
tf = vectorizer.fit_transform(df['cleaned'])

tfidf_vectorizer = TfidfVectorizer(max_df=0.95, min_df=2)
tfidf = tfidf_vectorizer.fit_transform(df['cleaned'])

# LDA
lda = LatentDirichletAllocation(n_components=10, random_state=42)
lda_topics = lda.fit_transform(tf)

```

```
# LSA
lsa = TruncatedSVD(n_components=10, random_state=42)
lsa_topics = lsa.fit_transform(tfidf)

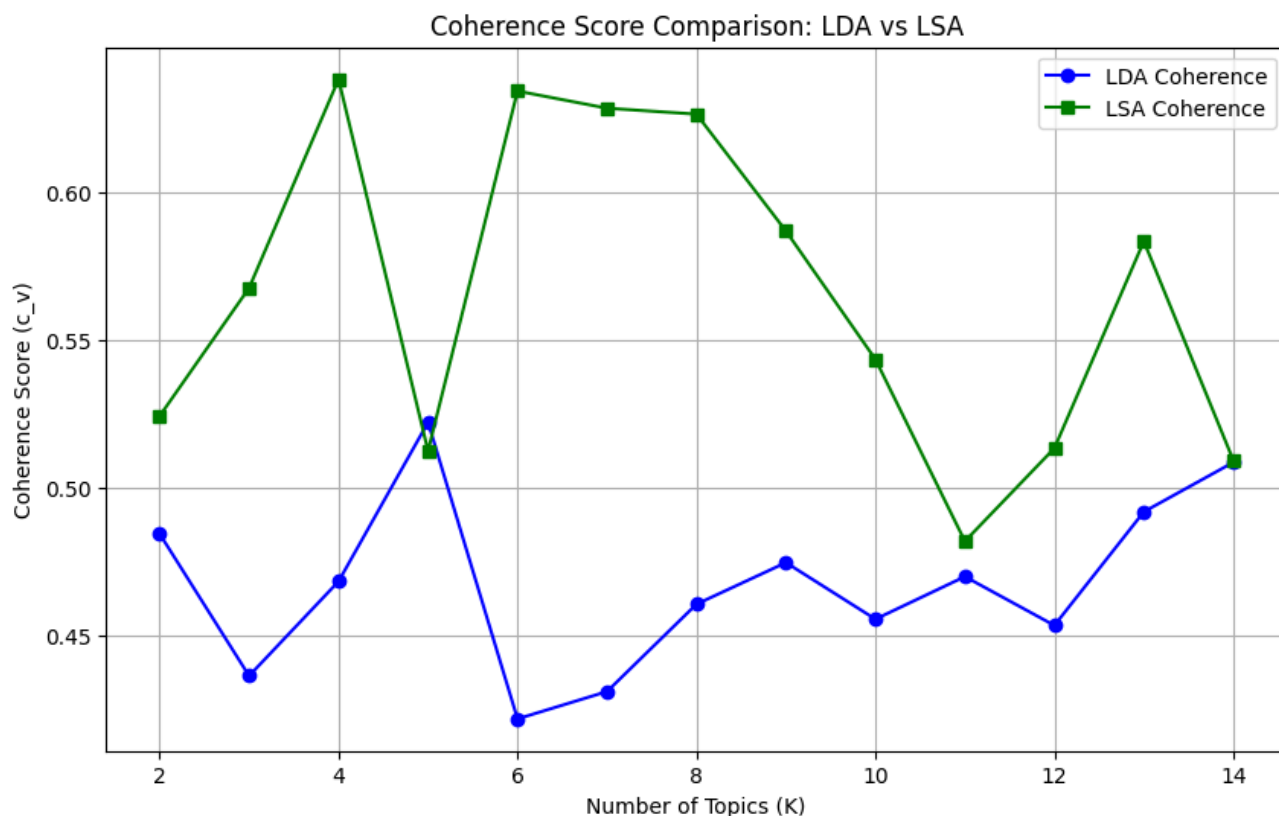
def compute_coherence_values(model_type, texts, dictionary, corpus, start=2, limit=15, step=1):
    coherence_scores = []
    for k in range(start, limit, step):
        if model_type == 'lda':
            model = gensim.models.LdaModel(corpus=corpus, id2word=dictionary, num_topics=
            elif model_type == 'lsa':
                model = gensim.models.LsiModel(corpus=corpus, id2word=dictionary, num_topics=
                coherencemodel = CoherenceModel(model=model, texts=texts, dictionary=dictionary,
                coherence_scores.append((k, coherencemodel.get_coherence()))
    return coherence_scores

lda_coherence = compute_coherence_values('lda', tokenized_docs, dictionary, corpus)
lsa_coherence = compute_coherence_values('lsa', tokenized_docs, dictionary, corpus)

import matplotlib.pyplot as plt

# Unpack the scores
lda_k, lda_scores = zip(*lda_coherence)
lsa_k, lsa_scores = zip(*lsa_coherence)

# Plotting
plt.figure(figsize=(10, 6))
plt.plot(lda_k, lda_scores, marker='o', label='LDA Coherence', color='blue')
plt.plot(lsa_k, lsa_scores, marker='s', label='LSA Coherence', color='green')
plt.xlabel("Number of Topics (K)")
plt.ylabel("Coherence Score (c_v)")
plt.title("Coherence Score Comparison: LDA vs LSA")
plt.legend()
plt.grid(True)
plt.show()
```



```
best_lda_model = gensim.models.LdaModel(corpus=corpus, id2word=dictionary, num_topics=5,

topics = best_lda_model.print_topics(num_words=10)
for idx, topic in topics:
    print(f"Topic {idx+1}: {topic}")
```



```
Topic 1: 0.005*"would" + 0.004*"also" + 0.004*"game" + 0.004*"like" + 0.004*"year" +
Topic 2: 0.010*"would" + 0.006*"people" + 0.005*"know" + 0.005*"think" + 0.004*"time"
Topic 3: 0.006*"stephanopoulos" + 0.005*"myers" + 0.004*"gopher" + 0.003*"president"
Topic 4: 0.004*"said" + 0.004*"know" + 0.004*"like" + 0.003*"would" + 0.003*"people"
Topic 5: 0.008*"file" + 0.004*"available" + 0.004*"window" + 0.004*"server" + 0.004*"
```



✓ BERTopic

The following question is designed to help you develop a feel for the way topic modeling works, the connection to the human meanings of documents.

Dataset from **assignment-3** (text dataset) .

Don't use any custom datasets.

Dataset must have 1000+ rows, no duplicates and null values

✓ Question 2 (20 Points)

Q2) Generate K=10 topics by using BERTopic and then find optimal K value by the coherence score. Interpret each topic and visualize with suitable style.

```
!pip install 'numpy>=1.24'
!pip install --upgrade jax bertopic
```

```

Requirement already satisfied: numpy>=1.24 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: jax in /usr/local/lib/python3.11/dist-packages (0.5
Requirement already satisfied: bertopic in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: jaxlib<=0.5.3,>=0.5.3 in /usr/local/lib/python3.11/
Requirement already satisfied: ml_dtypes>=0.4.0 in /usr/local/lib/python3.11/dist-
Collecting numpy>=1.25 (from jax)
  Using cached numpy-2.2.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.
Requirement already satisfied: opt_einsum in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: scipy>=1.11.1 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: hdbscan>=0.8.29 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: pandas>=1.1.5 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: scikit-learn>=1.0 in /usr/local/lib/python3.11/dist
Requirement already satisfied: sentence-transformers>=0.4.1 in /usr/local/lib/pyth
Requirement already satisfied: tqdm>=4.41.1 in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: umap-learn>=0.5.0 in /usr/local/lib/python3.11/dist
Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-package
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/d
Requirement already satisfied: transformers<5.0.0,>=4.41.0 in /usr/local/lib/pytho
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.1
Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.
Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python
  Using cached numpy-2.0.2-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.4.127 in /usr/local/lib/
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.4.127 in /usr/local/li
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.4.127 in /usr/local/lib/

```

```
Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in /usr/local/lib/python3.11/dist-packages/nvidia/cudnn/cu12
Requirement already satisfied: nvidia-cublas-cu12==12.4.5.8 in /usr/local/lib/python3.11/dist-packages/nvidia/cublas/cu12
Requirement already satisfied: nvidia-cufft-cu12==11.2.1.3 in /usr/local/lib/python3.11/dist-packages/nvidia/cufft/cu12
Requirement already satisfied: nvidia-curand-cu12==10.3.5.147 in /usr/local/lib/python3.11/dist-packages/nvidia/curand/cu12
Requirement already satisfied: nvidia-cusolver-cu12==11.6.1.9 in /usr/local/lib/python3.11/dist-packages/nvidia/cusolver/cu12
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages/nvidia/cusparselt/cu12
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages/nvidia/nccl/cu12
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages/nvidia/nvtx/cu12
Requirement already satisfied: nvidia-nvjitlink-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages/nvidia/nvjitlink/cu12
Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-packages/triton
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages/sympy
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages/mpmath
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist-packages/regex
```

```
!pip install --upgrade numpy --quiet
!pip uninstall -y bertopic
!pip install bertopic[all] --quiet
```

ERROR: pip's dependency resolver does not currently take into account all the package dependencies that were installed when a package was installed. You may consider using pip install --upgrade to update pip. (Use pip install --upgrade pip to install the latest pip.)

tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.2.4 which is incompatible.

gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.2.4 which is incompatible.

tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow 2.19.0 which is incompatible.

numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.2.4 which is incompatible.

tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which is incompatible.

Found existing installation: bertopic 0.17.0

Uninstalling bertopic-0.17.0:

Successfully uninstalled bertopic-0.17.0

WARNING: bertopic 0.17.0 does not provide the extra 'all'

ERROR: pip's dependency resolver does not currently take into account all the package dependencies that were installed when a package was installed. You may consider using pip install --upgrade to update pip. (Use pip install --upgrade pip to install the latest pip.)

gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.0.2 which is incompatible.

tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow 2.19.0 which is incompatible.

tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which is incompatible.

```
!pip install --upgrade numpy --quiet
!pip uninstall -y bertopic
!pip install bertopic[all] --quiet
!pip install --upgrade numpy --quiet
!pip uninstall -y umap-learn pynndescent
!pip install umap-learn[plot] pynndescent --no-cache-dir --quiet
!pip install --upgrade numpy --quiet
!pip uninstall -y pynndescent numpy
!pip install numpy==1.22.4
!pip install pynndescent
```




```
ERROR: pip's dependency resolver does not currently take into account all the pack
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.2.4 which is
gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.2.4 which is incompat
tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.2.4 which is incompat
tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 wh
Found existing installation: bertopic 0.17.0
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Uninstalling bertopic-0.17.0:
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Successfully uninstalled bertopic-0.17.0
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WARNING: bertopic 0.17.0 does not provide the extra 'all'
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ERROR: pip's dependency resolver does not currently take into account all the pack
gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.0.2 which is incompat
tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow
tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 wh
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ERROR: pip's dependency resolver does not currently take into account all the pack
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.2.4 which is
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tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.2.4 which is incompat
tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 wh
Found existing installation: umap-learn 0.5.7
```

```
Uninstalling umap-learn-0.5.7:
```

```
Successfully uninstalled umap-learn-0.5.7
```

```
Found existing installation: pynndescent 0.5.13
```

```
Uninstalling pynndescent-0.5.13:
```

```
Successfully uninstalled pynndescent-0.5.13
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```
19.5/19.5 MB 236.9 MB/s eta 0:00:00
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```
ERROR: pip's dependency resolver does not currently take into account all the pack
gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.0.2 which is incompat
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```
ERROR: pip's dependency resolver does not currently take into account all the pack
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.2.4 which is
gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.2.4 which is incompat
tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.2.4 which is incompat
tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 wh
Found existing installation: pynndescent 0.5.13
```

```
Uninstalling pynndescent-0.5.13:
```

```
Successfully uninstalled pynndescent-0.5.13
```

```
Found existing installation: numpy 2.2.4
```

```
Uninstalling numpy-2.2.4:
```

```
Successfully uninstalled numpy-2.2.4
```

```
Collecting numpy==1.22.4
```

```
Using cached numpy-1.22.4-cp311-cp311-linux_x86_64.whl
```

```
Installing collected packages: numpy
```

```
ERROR: pip's dependency resolver does not currently take into account all the pack
umap-learn 0.5.7 requires pynndescent>=0.5, which is not installed.
```

```
jaxlib 0.5.3 requires numpy>=1.25, but you have numpy 1.22.4 which is incompatible
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 1.22.4 which i
jax 0.5.3 requires numpy>=1.25, but you have numpy 1.22.4 which is incompatible.
```

```
ml-dtypes 0.5.1 requires numpy>=1.23.3; python_version >= "3.11", but you have num
cuml-cu12 25.2.1 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which is i
scikit-image 0.25.2 requires numpy>=1.24, but you have numpy 1.22.4 which is incomp
pylibraft-cu12 25.2.0 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which
dask-cudf-cu12 25.2.2 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which
mizani 0.13.2 requires numpy>=1.23.5, but you have numpy 1.22.4 which is incompati
astropy 7.0.1 requires numpy>=1.23.2, but you have numpy 1.22.4 which is incompati
pylibcugraph-cu12 25.2.0 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 wh
ucx-py-cu12 0.42.0 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which is
```



```

plotnine 0.14.5 requires numpy>=1.23.5, but you have numpy 1.22.4 which is incompatible
xarray 2025.1.2 requires numpy>=1.24, but you have numpy 1.22.4 which is incompatible
pandas 2.2.2 requires numpy>=1.23.2; python_version == "3.11", but you have numpy
pandas-stubs 2.2.2.240909 requires numpy>=1.23.5, but you have numpy 1.22.4 which
contourpy 1.3.1 requires numpy>=1.23, but you have numpy 1.22.4 which is incompatible
tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow
numexpr 2.10.2 requires numpy>=1.23.0, but you have numpy 1.22.4 which is incompatible
pymc 5.21.2 requires numpy>=1.25.0, but you have numpy 1.22.4 which is incompatible
treescope 0.1.9 requires numpy>=1.25.2, but you have numpy 1.22.4 which is incompatible
dm-tree 0.1.9 requires numpy>=1.23.3; python_version >= "3.11", but you have numpy
dask-cuda 25.2.0 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which is incompatible
cudf-cu12 25.2.1 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which is incompatible
arviz 0.21.0 requires numpy>=1.23.0, but you have numpy 1.22.4 which is incompatible
matplotlib 3.10.0 requires numpy>=1.23, but you have numpy 1.22.4 which is incompatible
cuvs-cu12 25.2.1 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which is incompatible
imbalanced-learn 0.13.0 requires numpy<3,>=1.24.3, but you have numpy 1.22.4 which
albumintations 2.0.5 requires numpy>=1.24.4, but you have numpy 1.22.4 which is incompatible
flax 0.10.5 requires numpy>=1.23.2; python_version >= "3.11", but you have numpy 1
ucxx-cu12 0.42.0 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which is incompatible
blosc2 3.2.1 requires numpy>=1.26, but you have numpy 1.22.4 which is incompatible
rmm-cu12 25.2.0 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which is incompatible
albucore 0.0.23 requires numpy>=1.24.4, but you have numpy 1.22.4 which is incompatible
chex 0.1.89 requires numpy>=1.24.1, but you have numpy 1.22.4 which is incompatible
opencv-python-headless 4.11.0.86 requires numpy>=1.23.5; python_version >= "3.11",
tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which
bigframes 1.42.0 requires numpy>=1.24.0, but you have numpy 1.22.4 which is incompatible
xarray-einstats 0.8.0 requires numpy>=1.23, but you have numpy 1.22.4 which is incompatible
opencv-python 4.11.0.86 requires numpy>=1.23.5; python_version >= "3.11", but you
nx-cugraph-cu12 25.2.0 requires numpy<3.0a0,>=1.23, but you have numpy 1.22.4 which
opencv-contrib-python 4.11.0.86 requires numpy>=1.23.5; python_version >= "3.11",
Successfully installed numpy-1.22.4

```

Collecting pynndescent

Using cached pynndescent-0.5.13-py3-none-any.whl.metadata (6.8 kB)

Requirement already satisfied: scikit-learn>=0.18 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: scipy>=1.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: llvmlite>=0.30 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: joblib>=0.11 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: numpy<2.1,>=1.22 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/dist-packages

Using cached pynndescent-0.5.13-py3-none-any.whl (56 kB)

Installing collected packages: pynndescent

Successfully installed pynndescent-0.5.13

Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (1

ERROR: Operation cancelled by user

^C

RuntimeError

Traceback (most recent call last)

RuntimeError: module was compiled against NumPy C-API version 0x10 (NumPy 1.23) but the running NumPy has C-API version 0xf. Check the section C-API incompatibility at the Troubleshooting ImportError section at <https://numpy.org/devdocs/user/troubleshooting-importerror.html#c-api-incompatibility> for indications on how to solve this problem.

ImportError

Traceback (most recent call last)

<ipython-input-1-da0c5390cb4d> in <cell line: 0>()

12 import numpy as np

13 import pandas as pd

---> 14 import matplotlib.pyplot as plt

15 # The issue is related to compatibility with jax, tensorflow, and transformers

16 # A combination of downgrading jax and ensuring appropriate transformers version often helps resolve this issue

1 frames

/usr/local/lib/python3.11/dist-packages/matplotlib/__init__.py in

_check_versions()

```
258     module = importlib.import_module(modname)
259     if parse_version(module.__version__) < parse_version(minver):
--> 260         raise ImportError(f"Matplotlib requires {modname}>={minver};
"
261                               f"you have {module.__version__}")
262
```

ImportError: Matplotlib requires numpy>=1.23; you have 1.22.4

NOTE: If your import is failing due to a missing package, you can manually install dependencies using either !pip or !apt.

To view examples of installing some common dependencies, click the

```
!pip install numpy==1.24.4 # Install a compatible numpy version
!pip install --upgrade pandas # Reinstall/upgrade pandas with compatible numpy version
```

 Requirement already satisfied: numpy==1.24.4 in /usr/local/lib/python3.11/dist-packages
 Requirement already satisfied: pandas in /usr/local/lib/python3.11/dist-packages (2.2.3)
 Collecting pandas
 Downloading pandas-2.2.3-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (12.1 MB)
 Requirement already satisfied: numpy>=1.23.2 in /usr/local/lib/python3.11/dist-packages (1.24.4)
 Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (2.9.0)
 Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (2024.1)
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (2024.1)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (1.17.0)
 Downloading pandas-2.2.3-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (12.1 MB) (13.1/13.1 MB 121.2 MB/s eta 0:00:00)
 Installing collected packages: pandas
 Attempting uninstall: pandas
 Found existing installation: pandas 2.2.2
 Uninstalling pandas-2.2.2:
 Successfully uninstalled pandas-2.2.2
 ERROR: pip's dependency resolver does not currently take into account all the package dependencies that were requested by google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3 which is incompatible.
 Successfully installed pandas-2.2.3

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
# The issue is related to compatibility with jax, tensorflow, and transformers
# A combination of downgrading jax and ensuring appropriate transformers version often helps
!pip install jax[cpu]==0.4.13 # Downgrade jax to a compatible version
!pip install --upgrade transformers # Update transformers for compatibility
!pip install --upgrade bertopic[all] # Update bertopic[all]
from bertopic import BERTopic
from gensim.models.coherencemodel import CoherenceModel
from gensim.corpora import Dictionary

k = 10
# Make sure the path to your CSV file is correct
df = pd.read_csv('/content/sample_data/Narrators_data_Cleaned.csv', usecols=['Processed_Details'])
# Access the 'Processed_Details' column using the correct name
details = df['Processed_Details'].to_list()
df.head()
```



```
Requirement already satisfied: jax==0.4.13 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: ml_dtypes>=0.1.0 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: numpy>=1.21 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: opt_einsum in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: scipy>=1.7 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: jaxlib==0.4.13 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: transformers in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: huggingface-hub<1.0,>=0.30.0 in /usr/local/lib/pyth
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.11
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dis
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/pythor
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dis
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dis
Requirement already satisfied: bertopic[all] in /usr/local/lib/python3.11/dist-pac
WARNING: bertopic 0.17.0 does not provide the extra 'all'
Requirement already satisfied: hdbscan>=0.8.29 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: pandas>=1.1.5 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: scikit-learn>=1.0 in /usr/local/lib/python3.11/dist
Requirement already satisfied: sentence-transformers>=0.4.1 in /usr/local/lib/pyth
Requirement already satisfied: tqdm>=4.41.1 in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: umap-learn>=0.5.0 in /usr/local/lib/python3.11/dist
Requirement already satisfied: scipy>=1.0 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-package
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/c
Requirement already satisfied: transformers<5.0.0,>=4.41.0 in /usr/local/lib/pythc
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.1
Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.
Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/pythor
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.4.127 in /usr/local/lib/
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.4.127 in /usr/local/li
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.4.127 in /usr/local/lib/
Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in /usr/local/lib/pythc
```

```

Requirement already satisfied: nvidia-cublas-cu12==12.4.5.8 in /usr/local/lib/python3.11/dist-packages/nvidia-cublas-cu12-12.4.5.8
Requirement already satisfied: nvidia-cufft-cu12==11.2.1.3 in /usr/local/lib/python3.11/dist-packages/nvidia-cufft-cu12-11.2.1.3
Requirement already satisfied: nvidia-curand-cu12==10.3.5.147 in /usr/local/lib/python3.11/dist-packages/nvidia-curand-cu12-10.3.5.147
Requirement already satisfied: nvidia-cusolver-cu12==11.6.1.9 in /usr/local/lib/python3.11/dist-packages/nvidia-cusolver-cu12-11.6.1.9
Requirement already satisfied: nvidia-cusparse-cu12==12.3.1.170 in /usr/local/lib/python3.11/dist-packages/nvidia-cusparse-cu12-12.3.1.170
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages/nvidia-cusparselt-cu12-0.6.2
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages/nvidia-nccl-cu12-2.21.5
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages/nvidia-nvtx-cu12-12.4.127
Requirement already satisfied: nvidia-nvjitlink-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages/nvidia-nvjitlink-cu12-12.4.127
Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-packages/triton-3.2.0
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages/sympy-1.13.1
Requirement already satisfied: mpmath<1.4, >=1.1.0 in /usr/local/lib/python3.11/dist-packages/mpmath-1.3.0
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist-packages/regex-2024.11.22
Requirement already satisfied: tokenizers<0.22, >=0.21 in /usr/local/lib/python3.11/dist-packages/tokenizers-0.21.0
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dist-packages/safetensors-0.5.2
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages/MarkupSafe-2.1.5
Requirement already satisfied: charset-normalizer<4, >=2 in /usr/local/lib/python3.11/dist-packages/charset-normalizer-3.3.0
Requirement already satisfied: idna<4, >=2.5 in /usr/local/lib/python3.11/dist-packages/idna-3.10
Requirement already satisfied: urllib3<3, >=1.21.1 in /usr/local/lib/python3.11/dist-packages/urllib3-2.2.3
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages/certifi-2024.12.14
ERROR:jax._src.xla_bridge:Jax plugin configuration error: Plugin module %s could not be loaded
Traceback (most recent call last):
  File "/usr/local/lib/python3.11/dist-packages/jax/_src/xla_bridge.py", line 420, in _load_plugin
    plugin_module = importlib.import_module(plugin_module_name)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.11/importlib/__init__.py", line 126, in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "<frozen importlib._bootstrap>", line 1204, in _gcd_import
  File "<frozen importlib._bootstrap>", line 1176, in _find_and_load
  File "<frozen importlib._bootstrap>", line 1147, in _find_and_load_unlocked
  File "<frozen importlib._bootstrap>", line 690, in _load_unlocked
  File "<frozen importlib._bootstrap_external>", line 940, in exec_module
  File "<frozen importlib._bootstrap>", line 241, in _call_with_frames_removed
  File "/usr/local/lib/python3.11/dist-packages/jax_plugins/xla_cuda12/__init__.py", line 1, in <module>
    from jax._src.lib import triton
ImportError: cannot import name 'triton' from 'jax._src.lib' (/usr/local/lib/python3.11/dist-packages/jax/_src/lib.py)

```

Processed_details

- 0 nisei femal born may selleck washington spent ...
- 1 nisei male born june seattl washington grew ar...
- 2 nisei femal born octob seattl washington famil...
- 3 nisei femal born juli boyl height california a...
- 4 sansei male born march torranc california grew...

```
Berttopic_model = BERTopic(nr_topics=k)
```

```
topics, probabilities = Berttopic_model.fit_transform(details)
```

```
Berttopic_model.get_topic_info()
```

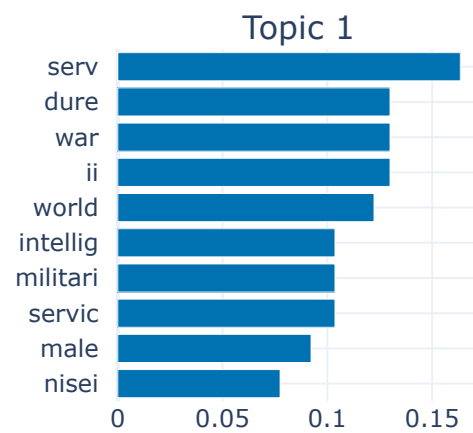
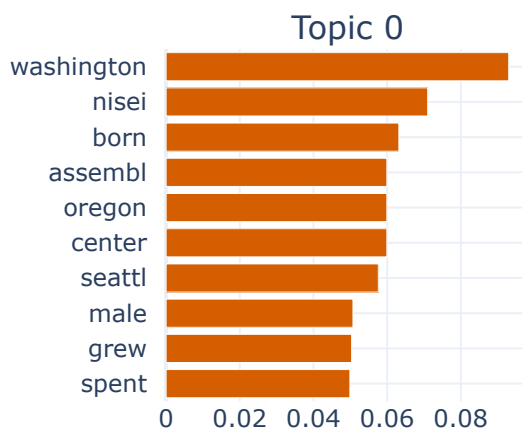


	Topic	Count	Name	Representation	Representative_Doc
0	0	410	0_washington_nisei_born_assembl	[washington, nisei, born, assembl, oregon, cen...	[nisei male bc septemb sea washington spe
1	1	164	1_serv_dure_war_ii	[serv, dure, war, ii, world, intellig, militar...	[nisei male born j dure world war ii se r
2	2	123	2_pearl_bomb_california_juli	[pearl, bomb, california, juli, famili, went, ...	[nisei male born j lindsay california follo
3	3	82	3_seattl_white_district_reverend	[seattl, white, district, reverend, novemb, mo...	[white male bc novemb sea washington son
				[success. prior.	[nisei femal born oct

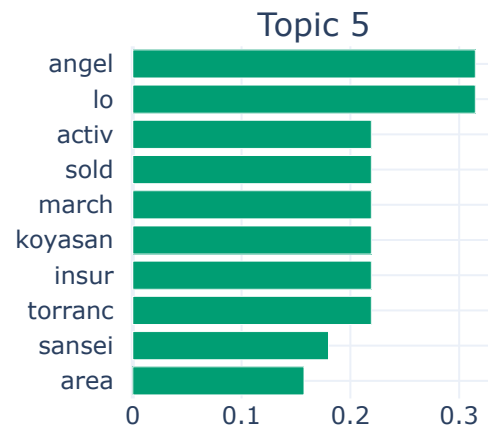
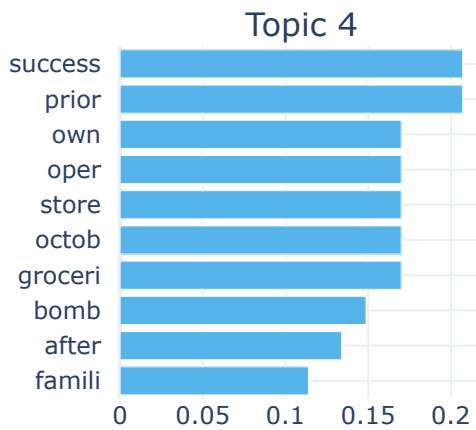
```
Berttopic_model.visualize_barchart(top_n_topics=10, n_words = 40, width = 300, height = 3
```



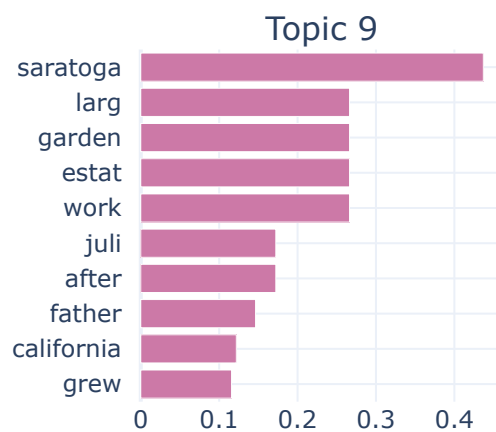
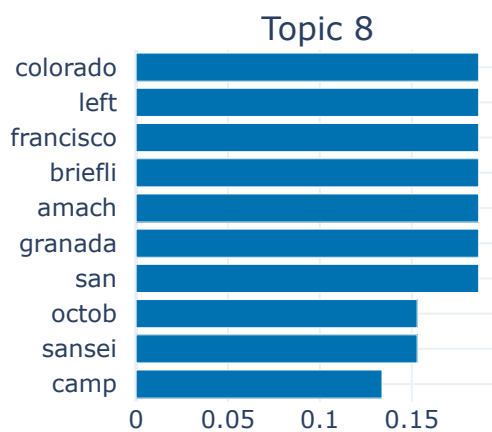
Topic Word Sc



pearl
bomb
california
juli
famili
went
height
harbor
lindsay
gardena



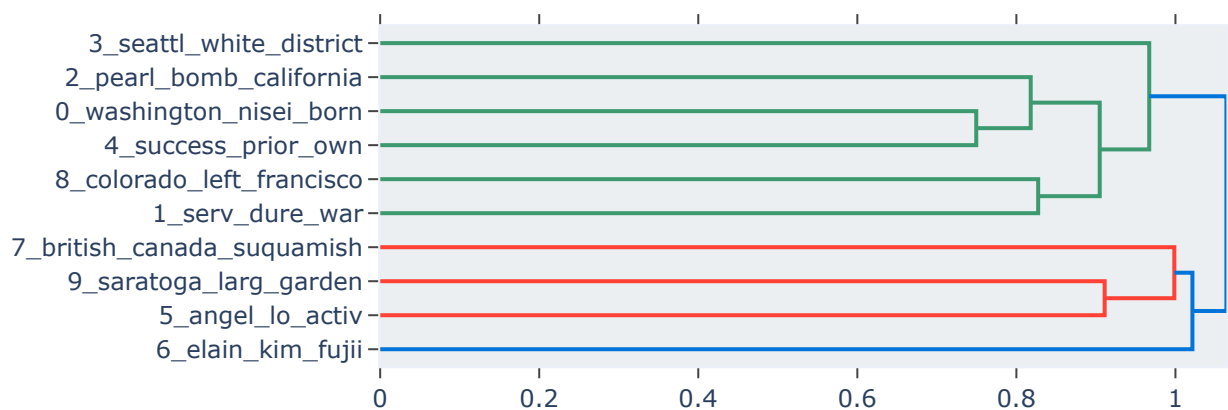
elain
kim
fujii
arlen
ann
akagi
yoshitomi
karen
bill
lindwal



`Berttopic_model.visualize_hierarchy(top_n_topics=10, width = 700, height = 700)`



Hierarchical Clustering



```
pip install gensim
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.11/dist-packages (4.3.3)
Requirement already satisfied: numpy<2.0, >=1.18.5 in /usr/local/lib/python3.11/dist-packages (1.24.4)
Requirement already satisfied: scipy<1.14.0, >=1.7.0 in /usr/local/lib/python3.11/dist-packages (1.10.1)
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.11/dist-packages (7.0.5)
Requirement already satisfied: wrapt in /usr/local/lib/python3.11/dist-packages (from gensim) (1.16.0)
```

```
!pip install --upgrade h5py
```

```
Requirement already satisfied: h5py in /usr/local/lib/python3.11/dist-packages (3.13.0)
Requirement already satisfied: numpy>=1.19.3 in /usr/local/lib/python3.11/dist-packages (1.24.4)
```

```
!pip install --upgrade jax jaxlib
```

```
Requirement already satisfied: jax in /usr/local/lib/python3.11/dist-packages (0.4.13)
Collecting jax
  Using cached jax-0.5.3-py3-none-any.whl.metadata (22 kB)
Requirement already satisfied: jaxlib in /usr/local/lib/python3.11/dist-packages (0.4.13)
Collecting jaxlib
  Using cached jaxlib-0.5.3-cp311-cp311-manylinux2014_x86_64.whl.metadata (1.2 kB)
Requirement already satisfied: ml_dtypes>=0.4.0 in /usr/local/lib/python3.11/dist-packages (0.4.0)
Collecting numpy>=1.25 (from jax)
  Using cached numpy-2.2.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (17.1 MB)
Requirement already satisfied: opt_einsum in /usr/local/lib/python3.11/dist-packages (from jax) (3.4.0)
Requirement already satisfied: scipy>=1.11.1 in /usr/local/lib/python3.11/dist-packages (from jax) (1.10.1)
Using cached jax-0.5.3-py3-none-any.whl (2.4 MB)
Using cached jaxlib-0.5.3-cp311-cp311-manylinux2014_x86_64.whl (105.1 MB)
Using cached numpy-2.2.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (17.1 MB)
Installing collected packages: numpy, jaxlib, jax
  Attempting uninstall: numpy
    Found existing installation: numpy 1.24.4
    Uninstalling numpy-1.24.4:
```

```

Successfully uninstalled numpy-1.24.4
Attempting uninstall: jaxlib
Found existing installation: jaxlib 0.4.13
Uninstalling jaxlib-0.4.13:
Successfully uninstalled jaxlib-0.4.13
Attempting uninstall: jax
Found existing installation: jax 0.4.13
Uninstalling jax-0.4.13:
Successfully uninstalled jax-0.4.13
ERROR: pip's dependency resolver does not currently take into account all the package
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.2.4 which is in
gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.2.4 which is incompati
google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3 which is incompa
tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow 2.1
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.2.4 which is incompatibl
tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which
Successfully installed jax-0.5.3 jaxlib-0.5.3 numpy-2.2.4

```

```

!pip install numpy==1.24.4 # Install a compatible numpy version
# Reinstall/upgrade gensim with compatible numpy version

```

```

⇒ Requirement already satisfied: numpy==1.24.4 in /usr/local/lib/python3.11/dist-packag
◀────────────────────────────────────────────────────────────────────────────────▶

```

```
!pip install --upgrade gensim
```

```

⇒ Requirement already satisfied: gensim in /usr/local/lib/python3.11/dist-packages (4.3
Requirement already satisfied: numpy<2.0,>=1.18.5 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: scipy<1.14.0,>=1.7.0 in /usr/local/lib/python3.11/dist
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: wrapt in /usr/local/lib/python3.11/dist-packages (from
◀────────────────────────────────────────────────────────────────────────────────▶
◀────────────────────────────────────────────────────────────────────────────────▶

```

```
!pip install numpy==1.24.4 # Install a compatible numpy version
```

```

⇒ Requirement already satisfied: numpy==1.24.4 in /usr/local/lib/python3.11/dist-packag
◀────────────────────────────────────────────────────────────────────────────────▶

```

```

# Install a compatible numpy version
!pip install --upgrade jax jaxlib # Upgrade jax and jaxlib for compatibility
#!pip install --upgrade bertopic[all] # Update bertopic[all] for compatibility with lates

```

```

⇒ Requirement already satisfied: jax in /usr/local/lib/python3.11/dist-packages (0.5.3)
Requirement already satisfied: jaxlib in /usr/local/lib/python3.11/dist-packages (0.5
Requirement already satisfied: ml_dtypes>=0.4.0 in /usr/local/lib/python3.11/dist-pac
Collecting numpy>=1.25 (from jax)
Using cached numpy-2.2.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
Requirement already satisfied: opt_einsum in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: scipy>=1.11.1 in /usr/local/lib/python3.11/dist-packag
Using cached numpy-2.2.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (
Installing collected packages: numpy
Attempting uninstall: numpy

```

Found existing installation: numpy 1.24.4

Uninstalling numpy-1.24.4:

Successfully uninstalled numpy-1.24.4

ERROR: pip's dependency resolver does not currently take into account all the package dependencies that were installed when you created this environment. Here are the package dependencies that were installed when you created this environment, and the versions that were installed:

- tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.2.4 which is incompatible
- gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.2.4 which is incompatible
- google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3 which is incompatible
- tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow 2.19.0 which is incompatible
- numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.2.4 which is incompatible
- tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which is incompatible

Successfully installed numpy-2.2.4

```
#!pip install numpy==1.24.4 # Install a compatible numpy version
```

```
#!pip install --upgrade jax jaxlib # Upgrade jax and jaxlib for compatibility
```

```
!pip install --upgrade bertopic[all] # Update bertopic[all] for compatibility with latest
```

Requirement already satisfied: bertopic[all] in /usr/local/lib/python3.11/dist-packages
WARNING: bertopic 0.17.0 does not provide the extra 'all'

Requirement already satisfied: hdbscan>=0.8.29 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: pandas>=1.1.5 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: scikit-learn>=1.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: sentence-transformers>=0.4.1 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: tqdm>=4.41.1 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: umap-learn>=0.5.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: scipy>=1.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: transformers<5.0.0,>=4.41.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.11/dist-packages

Collecting numpy>=1.20.0 (from bertopic[all])

Using cached numpy-2.0.2-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: nvidia-cublas-cu12==12.4.5.8 in /usr/local/lib/python3.11/dist-packages

Requirement already satisfied: nvidia-cufft-cu12==11.2.1.3 in /usr/local/lib/python3.11/dist-packages

```
Requirement already satisfied: nvidia-curand-cu12==10.3.5.147 in /usr/local/lib/py
Requirement already satisfied: nvidia-cusolver-cu12==11.6.1.9 in /usr/local/lib/py
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/pyt
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python
Requirement already satisfied: nvidia-nvjitlink-cu12==12.4.127 in /usr/local/lib/p
Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dis
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist
Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.11
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dis
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.
```

```
!pip install numpy==1.24.4 # Install a compatible numpy version
```

```
#!pip install --upgrade gensim # Reinstall/upgrade gensim with compatible numpy version
```



```
Collecting numpy==1.24.4
```

```
Using cached numpy-1.24.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
```

```
Using cached numpy-1.24.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
```

```
Installing collected packages: numpy
```

```
Attempting uninstall: numpy
```

```
Found existing installation: numpy 2.0.2
```

```
Uninstalling numpy-2.0.2:
```

```
Successfully uninstalled numpy-2.0.2
```

```
ERROR: pip's dependency resolver does not currently take into account all the package
jaxlib 0.5.3 requires numpy>=1.25, but you have numpy 1.24.4 which is incompatible.
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 1.24.4 which is i
jax 0.5.3 requires numpy>=1.25, but you have numpy 1.24.4 which is incompatible.
google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3 which is incompa
tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow 2.1
pymc 5.21.2 requires numpy>=1.25.0, but you have numpy 1.24.4 which is incompatible.
treescope 0.1.9 requires numpy>=1.25.2, but you have numpy 1.24.4 which is incompatib
blosc2 3.2.1 requires numpy>=1.26, but you have numpy 1.24.4 which is incompatible.
tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which
Successfully installed numpy-1.24.4
```

```
!pip install --upgrade gensim
```



```
Requirement already satisfied: gensim in /usr/local/lib/python3.11/dist-packages (4.3
```

```
Requirement already satisfied: numpy<2.0,>=1.18.5 in /usr/local/lib/python3.11/dist-p
```

```
Requirement already satisfied: scipy<1.14.0,>=1.7.0 in /usr/local/lib/python3.11/dist
```

```
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.11/dist-pa
```

```
Requirement already satisfied: wrapt in /usr/local/lib/python3.11/dist-packages (from
```

```
from gensim.models import CoherenceModel
from gensim.corpora import Dictionary
from bertopic import BERTopic
```

```
import pandas as pd
k = 10
# Make sure the path to your CSV file is correct
df = pd.read_csv('/content/sample_data/Narrators_data_Cleaned.csv', usecols=['Processed_d
# Access the 'Processed_Details' column using the correct name
details = df['Processed_details'].to_list() # Define details here
df.head()
```



Processed_details

- | | Processed_details |
|---|---|
| 0 | nisei femal born may selleck washington spent ... |
| 1 | nisei male born june seattl washington grew ar... |
| 2 | nisei femal born octob seattl washington famil... |
| 3 | nisei femal born juli boyl height california a... |
| 4 | sansei male born march torranc california grew... |

Start coding or [generate](#) with AI.

```
#from gensim.corpora import Dictionaryfrom bertopic import BERTopic # Ensure Dictionary i
def calculate_coherence_score(docs, min_topics=2, max_topics=10):
    coherence_scores = []

    for num_topics in range(min_topics, max_topics + 1):

        topic_model = BERTopic(nr_topics=num_topics)
        topics, _ = topic_model.fit_transform(docs)

        topic_keywords = [
            [word for word, _ in topic_model.get_topic(topic)]
            for topic in topic_model.get_topics().keys()
            if topic != -1
        ]

        tokenized_docs = [doc.split() for doc in docs]
        dictionary = Dictionary(tokenized_docs)

        coherence_model = CoherenceModel(
            topics=topic_keywords,
            dictionary=dictionary,
            texts=tokenized_docs,
            coherence='c_v'
        )
        score = coherence_model.get_coherence()
        coherence_scores.append((num_topics, score))
```

```
print(f"Topics={num_topics}, Coherence Score={score:.4f}")
```

```
return coherence_scores
```

```
coherence_scores = calculate_coherence_score(details, min_topics=2, max_topics=20) # Now
```

```

↳ Topics=2, Coherence Score=0.6088
Topics=3, Coherence Score=0.6288
Topics=4, Coherence Score=0.5740
Topics=5, Coherence Score=0.6748
Topics=6, Coherence Score=0.7004
Topics=7, Coherence Score=0.7599
Topics=8, Coherence Score=0.7813
Topics=9, Coherence Score=0.7575
Topics=10, Coherence Score=0.8184
Topics=11, Coherence Score=0.8044
Topics=12, Coherence Score=0.8492
Topics=13, Coherence Score=0.8659
Topics=14, Coherence Score=0.8955
Topics=15, Coherence Score=0.8998
Topics=16, Coherence Score=0.9241
Topics=17, Coherence Score=0.9072
Topics=18, Coherence Score=0.9214
Topics=19, Coherence Score=0.9241
Topics=20, Coherence Score=0.8755

```

```
best_topic_count = 10
```

```
final_model = BERTopic(nr_topics=best_topic_count)
```

```
final_topics, final_probs = final_model.fit_transform(details)
```

```
def evaluate_coherence(documents, min_topics=2, max_topics=20):
```

```
    scores = []
```

```
    for num_topics in range(min_topics, max_topics + 1):
```

```
        topic_model = BERTopic(nr_topics=num_topics)
```

```
        _, _ = topic_model.fit_transform(documents)
```

```
        topic_terms = [list(dict(topic_model.get_topic(i)).keys()) for i in range(num_top
```

```
        # Create dictionary and corpus for coherence calculation
```

```
        dictionary = Dictionary([terms for terms in topic_terms])
```

```
        corpus = [dictionary.doc2bow(terms) for terms in topic_terms]
```

```
        # Calculate coherence score using the c_v metric
```

```
        coherence_model = CoherenceModel(
```

```
            topics=topic_terms,
```

```
            texts=[doc.split() for doc in documents],
```

```
            dictionary=dictionary,
```

```
            coherence='c_v'
```

```
        )
```

```
        scores.append((num_topics, coherence_model.get_coherence()))
```

```
return scores
```

```
model = BERTopic(nr_topics=best_topic_count)
topic_results, topic_probabilities = model.fit_transform(details)

# Interpret the topics by examining their top words
print("\nTopic Interpretation (Top Words):")
for topic_num in range(best_topic_count):
    print(f"Topic {topic_num}:")
    print(model.get_topic(topic_num))
    print("\n")

# Visualize the topics
model.visualize_topics()
model.visualize_barchart(top_n_topics=12, n_words=10, width=350, height=350)
model.visualize_hierarchy(top_n_topics=12, width=700, height=700)
```



Topic Interpretation (Top Words):

Topic 0:

[('washington', 0.09296766098183444), ('nisei', 0.07099198915447157), ('born', 0.0

Topic 1:

[('serv', 0.16346517452134893), ('dure', 0.12994000846532136), ('war', 0.129940008

Topic 2:

[('pearl', 0.12743181204730392), ('bomb', 0.11150454832445877), ('california', 0.1

Topic 3:

[('seattl', 0.18731427296699393), ('white', 0.18535536297789665), ('district', 0.1

Topic 4:

[('success', 0.20709712905629496), ('prior', 0.20709712905629496), ('own', 0.16990

Topic 5:

[('angel', 0.3148363717396483), ('lo', 0.3148363717396483), ('activ', 0.2192793131

Topic 6:

[('elain', 0.32193299885634674), ('kim', 0.19619728015859522), ('fujii', 0.1961972

Topic 7:

[('british', 0.21927931311842996), ('canada', 0.21927931311842996), ('suquamish',

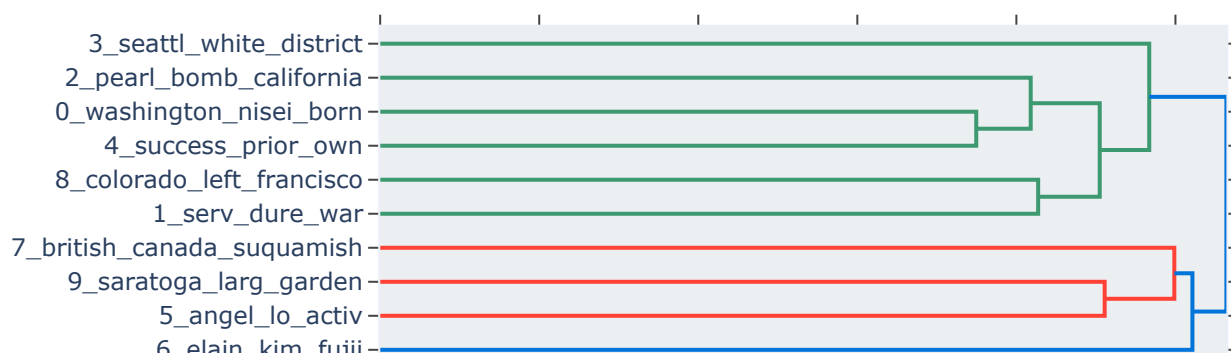
Topic 8:

[('colorado', 0.18638741615066548), ('left', 0.18638741615066548), ('francisco', 0

Topic 9:

[('saratoga', 0.4369090698764706), ('larg', 0.2662677373580935), ('garden', 0.2662

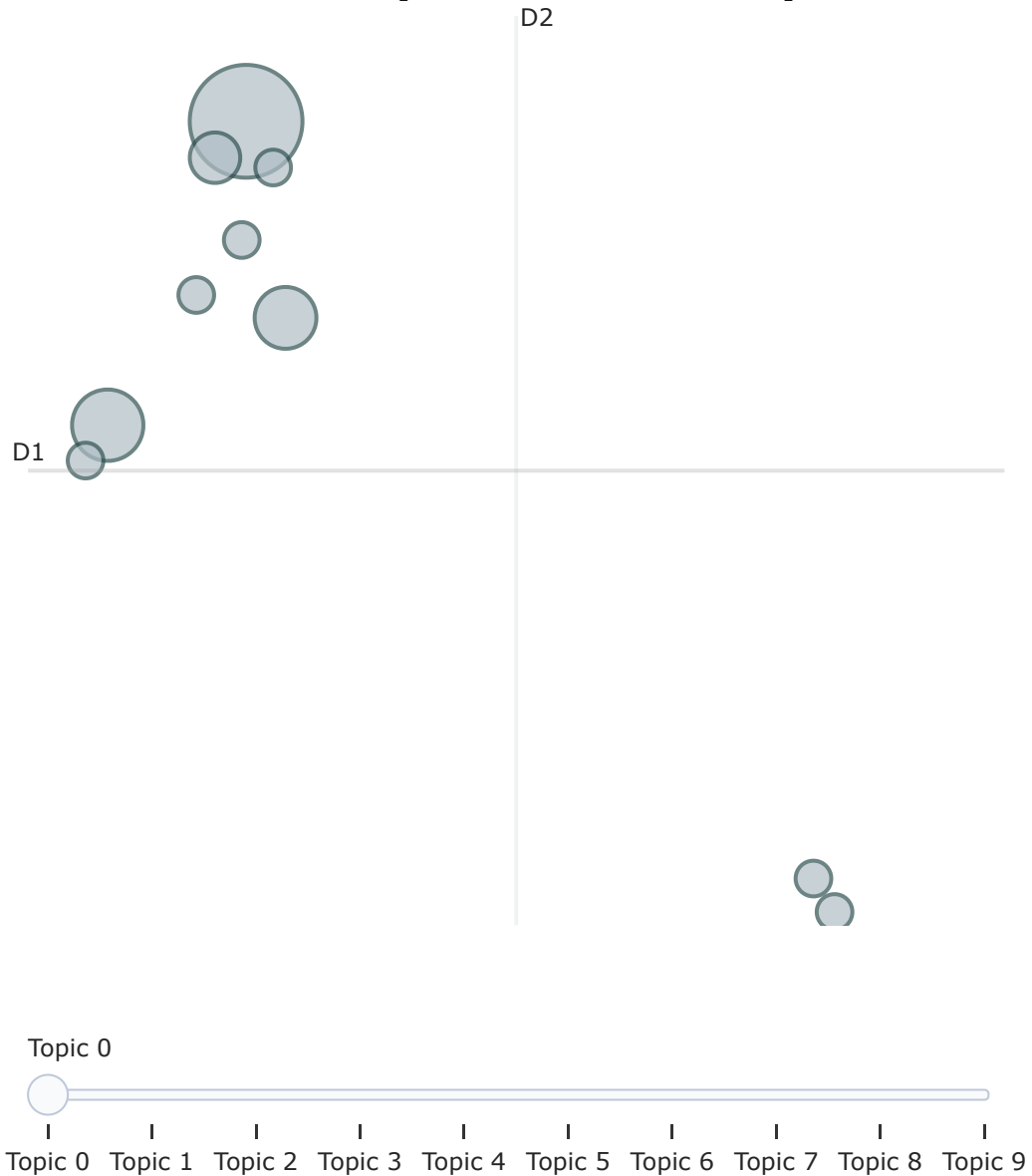
Hierarchical Clustering



model.visualize_topics()



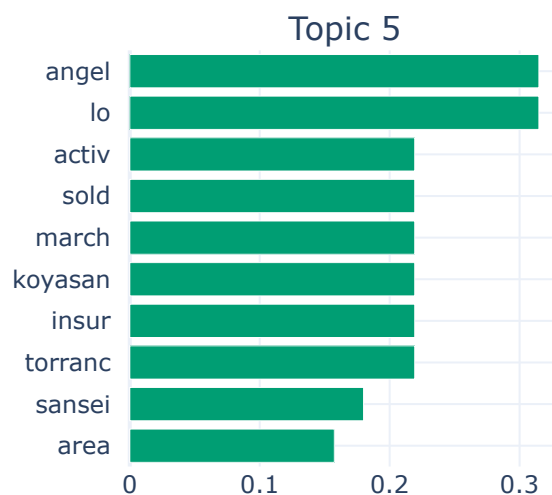
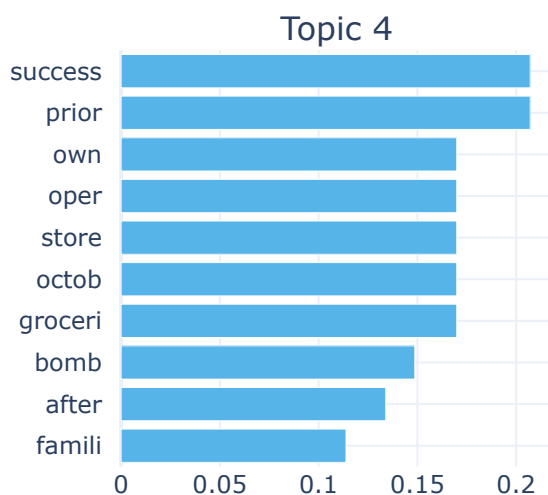
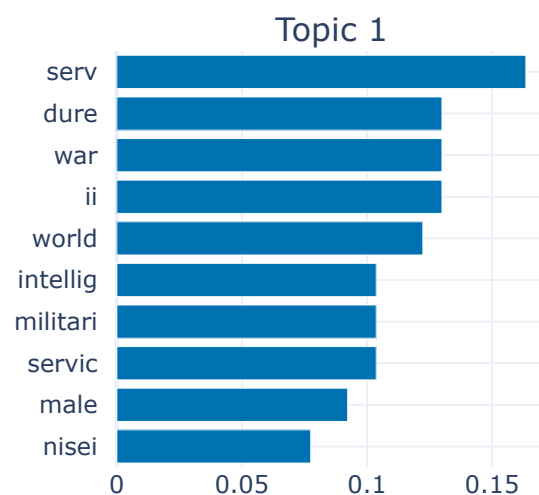
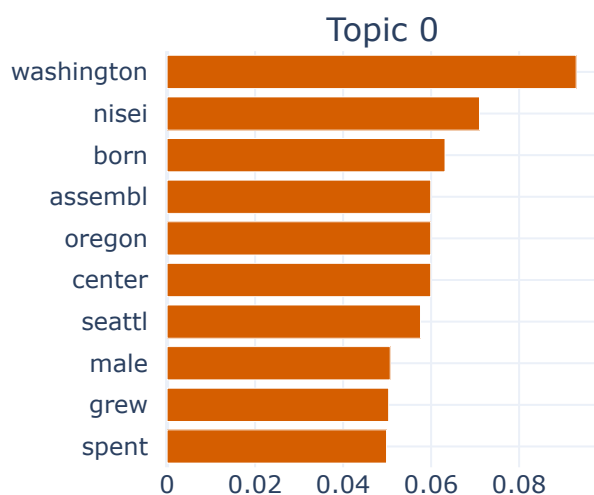
Intertopic Distance Map



```
model.visualize_barchart(top_n_topics=8, n_words = 10, width = 350, height = 350)
```



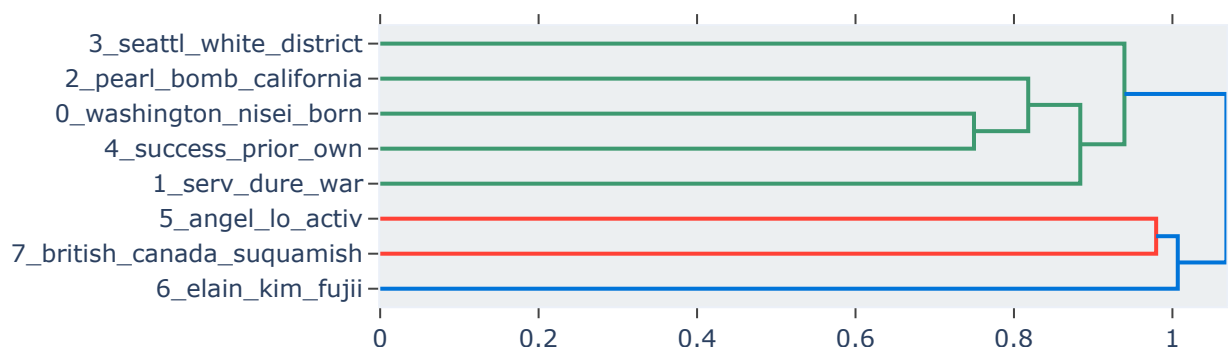
Topic



```
model.visualize_hierarchy(top_n_topics=8, width = 700, height = 700)
```



Hierarchical Clustering



✓ Question 3 (25 points)

Dataset Link: 20 Newsgroup Dataset (Random 2000 values)

Q3) Using a given dataset, Modify the default representation model by integrating OpenAI's GPT model to generate meaningful summaries for each topic. Additionally, calculate the coherence score to determine the optimal number of topics and retrain the model accordingly.

Usefull Link:

https://maartengr.github.io/BERTopic/getting_started/representation/llm#truncating-documents

```
import pandas as pd
import random
from sklearn.datasets import fetch_20newsgroups

# Load dataset and sample 2000 rows
data = fetch_20newsgroups(subset='all', remove=('headers', 'footers', 'quotes'))
sampled_data = random.sample(data.data, 2000)

# Convert to DataFrame
df = pd.DataFrame(sampled_data, columns=['text'])
print(df.head())
```



```
text
0  \n\n\na 286 upgrade would probably cost about ...
1  I've just managed to get xdm running from an N...
2  \nThe thing is, I know what arms and legs are....
3  \n\n\nWe use it as starting fluid for jet ski'...
4  \nI'll tell you all that I know about chromium...
```

```

import re
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

# Download necessary NLTK resources (run once)
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('punkt_tab') # Download the missing punkt_tab data

# Preprocessing tools
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()

# Preprocessing function
def preprocess(text):
    text = text.lower()
    text = re.sub(r'^a-z\s', '', text)
    tokens = nltk.word_tokenize(text)
    tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in stop_words]
    return " ".join(tokens)

# Apply preprocessing
df['cleaned'] = df['text'].apply(preprocess)
print(df[['text', 'cleaned']].head())

```

```

[>>> [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!

```

```

text \
0  \n\n\na 286 upgrade would probably cost about ...
1  I've just managed to get xdm running from an N...
2  \nThe thing is, I know what arms and legs are....
3  \n\n\nWe use it as starting fluid for jet ski'...
4  \nI'll tell you all that I know about chromium...

```

```

cleaned
0  upgrade would probably cost coprocessors accel...
1  managed running running xfree xdisplay pretty ...
2  thing know arm leg therefore generally easy te...
3  starting fluid ski fact lubricant work well si...
4  tell know chromium want thing chest blasted em...

```

```

!pip install numpy==1.24.4 # Install a compatible numpy version
#!pip install --upgrade gensim # Reinstall/upgrade gensim with compatible numpy version

```

```

Collecting numpy==1.24.4
  Using cached numpy-1.24.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
Using cached numpy-1.24.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
Installing collected packages: numpy
  Attempting uninstall: numpy
    Found existing installation: numpy 2.0.2
    Uninstalling numpy-2.0.2:
      Successfully uninstalled numpy-2.0.2
ERROR: pip's dependency resolver does not currently take into account all the package
jaxlib 0.5.3 requires numpy>=1.25, but you have numpy 1.24.4 which is incompatible.
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 1.24.4 which is i
jax 0.5.3 requires numpy>=1.25, but you have numpy 1.24.4 which is incompatible.
google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3 which is incompa
tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow 2.1
pymc 5.21.2 requires numpy>=1.25.0, but you have numpy 1.24.4 which is incompatible.
treescope 0.1.9 requires numpy>=1.25.2, but you have numpy 1.24.4 which is incompatib
blosc2 3.2.1 requires numpy>=1.26, but you have numpy 1.24.4 which is incompatible.
tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which
Successfully installed numpy-1.24.4

```

```

#!pip install numpy==1.24.4 # Install a compatible numpy version
!pip install --upgrade gensim # Reinstall/upgrade gensim with compatible numpy version#

```

```

Requirement already satisfied: gensim in /usr/local/lib/python3.11/dist-packages (4.3
Requirement already satisfied: numpy<2.0,>=1.18.5 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: scipy<1.14.0,>=1.7.0 in /usr/local/lib/python3.11/dist
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: wrapt in /usr/local/lib/python3.11/dist-packages (from

```

```
from gensim import corpora
```

```
# Tokenize preprocessed text
```

```
texts = [doc.split() for doc in df['cleaned']]
```

```
# Create dictionary and corpus
```

```
dictionary = corpora.Dictionary(texts)
```

```
corpus = [dictionary.doc2bow(text) for text in texts]
```

```
print(f"Sample dictionary tokens: {dictionary.token2id}")
```

```
print(f"Sample corpus: {corpus[0][:20]}")
```

```
#corpus
```

```

Sample dictionary tokens: {'accelerator': 0, 'card': 1, 'coprocessors': 2, 'cost': 3,
Sample corpus: [(0, 1), (1, 1), (2, 1), (3, 2), (4, 1), (5, 1), (6, 1), (7, 1), (8, 2

```

```
pip install numpy==1.24.4
```

Requirement already satisfied: numpy==1.24.4 in /usr/local/lib/python3.11/dist-packag

```
from gensim.models import LdaModel, CoherenceModel
import matplotlib.pyplot as plt
```

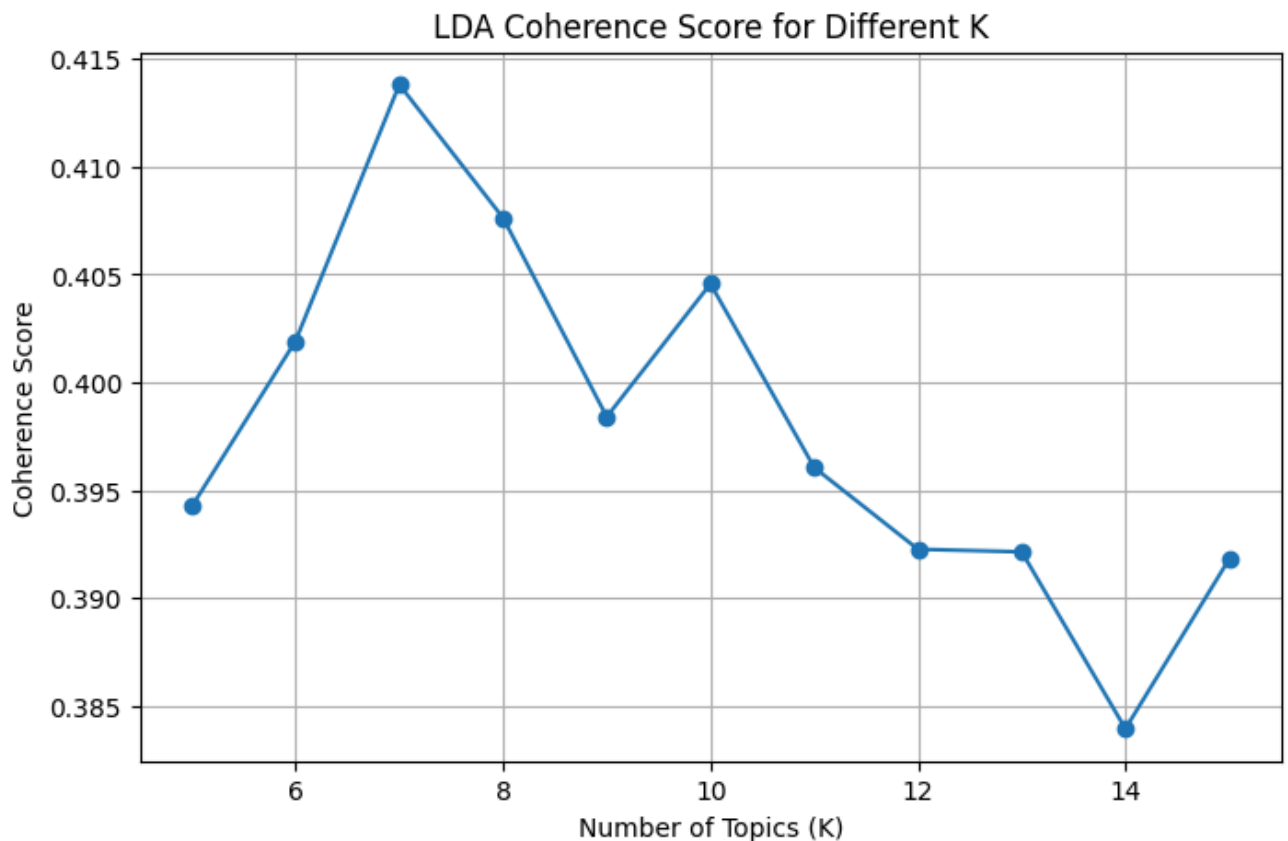
```
coherence_scores = []
```

```
for k in range(5, 16):
    lda = LdaModel(corpus=corpus, id2word=dictionary, num_topics=k, random_state=42)
    cm = CoherenceModel(model=lda, texts=texts, dictionary=dictionary, coherence='c_v')
    coherence = cm.get_coherence()
    coherence_scores.append((k, coherence))
    print(f"K={k}, Coherence Score={coherence:.4f}")
```

```
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=5, Coherence Score=0.3943
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=6, Coherence Score=0.4019
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=7, Coherence Score=0.4138
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=8, Coherence Score=0.4076
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=9, Coherence Score=0.3984
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=10, Coherence Score=0.4046
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=11, Coherence Score=0.3961
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=12, Coherence Score=0.3923
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=13, Coherence Score=0.3921
WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
K=14, Coherence Score=0.3839
K=15, Coherence Score=0.3918
```

```
# Plot coherence scores
k_vals, scores = zip(*coherence_scores)
plt.figure(figsize=(8, 5))
plt.plot(k_vals, scores, marker='o')
plt.xlabel("Number of Topics (K)")
plt.ylabel("Coherence Score")
plt.title("LDA Coherence Score for Different K")
plt.grid(True)
plt.show()

# Find best K
best_k = max(coherence_scores, key=lambda x: x[1])[0]
print(f"\nBest K based on coherence: {best_k}")
```



Best K based on coherence: 7

```
# Train LDA model with best K
lda_model = LdaModel(corpus=corpus, id2word=dictionary, num_topics=best_k, random_state=4)

# Print top keywords for each topic
topics = lda_model.show_topics(num_topics=best_k, num_words=10, formatted=False)

for idx, topic in topics:
    keywords = [word for word, prob in topic]
    print(f"Topic {idx+1}: {' '.join(keywords)}")
```

```

⏏ WARNING:gensim.models.ldamodel:too few updates, training might not converge; consider
Topic 1: would, know, people, like, well, time, thing, right, dont, year
Topic 2: people, would, dont, work, time, even, make, like, problem, could
Topic 3: would, like, dont, time, people, know, year, file, think, also
Topic 4: would, file, people, think, thing, know, dont, also, like, system
Topic 5: would, people, know, also, like, problem, dont, time, right, think
Topic 6: would, dont, know, people, file, like, also, want, time, right
Topic 7: dont, people, would, think, time, like, well, even, need, know

```

```
!pip install openai==0.27.8 # Install the specific older version
```

```

⏏ Collecting openai==0.27.8
  Using cached openai-0.27.8-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: requests>=2.20 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from
Requirement already satisfied: aiohttp in /usr/local/lib/python3.11/dist-packages (fr
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-package
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.11/d
Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-pa
Using cached openai-0.27.8-py3-none-any.whl (73 kB)
Installing collected packages: openai
  Attempting uninstall: openai
    Found existing installation: openai 1.71.0
    Uninstalling openai-1.71.0:
      Successfully uninstalled openai-1.71.0

```

```
import openai
```

```
openai.api_key = "sk-proj-8XLwzpt8YJERDPXpEfWKRblsUqnVttRB7ZK76BEMg2eIEoxNAZs2Dhm-qgKZs9d"
```

```

def gpt_topic_summary(keywords):
    prompt = f"Generate a short, meaningful summary for a topic based on these keywords:
    response = openai.ChatCompletion.create(
        model="gpt-3.5-turbo",
        messages=[{"role": "user", "content": prompt}],
        max_tokens=50
    )
    return response.choices[0].message.content.strip()

```



```
# Generate summaries
print(" GPT Summaries ")
for idx, topic in topics:
    keywords = [word for word, prob in topic]
    summary = gpt_topic_summary(keywords)
    print(f"Topic {idx+1}: {summary}")
```



GPT Summaries

```
-----
RateLimitError                                Traceback (most recent call last)
<ipython-input-17-24a4427dcd4a> in <cell line: 0>()
    16 for idx, topic in topics:
    17     keywords = [word for word, prob in topic]
--> 18     summary = gpt_topic_summary(keywords)
    19     print(f"Topic {idx+1}: {summary}")

----- 5 frames -----
/usr/local/lib/python3.11/dist-packages/openai/api_requestor.py in
_interpret_response_line(self, rbody, rcode, rheaders, stream)
    761         stream_error = stream and "error" in resp.data
    762         if stream_error or not 200 <= rcode < 300:
--> 763             raise self.handle_error_response(
    764                 rbody, rcode, resp.data, rheaders, stream_error=stream_error
    765             )
```

RateLimitError: You exceeded your current quota, please check your plan and billing details. For more information on this error, read the docs:
<https://platform.openai.com/docs/guides/error-codes/api-errors>.

✓ Question 4 (35 Points)

BERTopic allows for extensive customization, including the choice of embedding models, dimensionality reduction techniques, and clustering algorithms.

Dataset Link: 20 Newsgroup Dataset (Random 2000 values)

4)

4.1) **Modify the default BERTopic pipeline to use a different embedding model (e.g., Sentence-Transformers) and a different clustering algorithm (e.g., DBSCAN instead of HDBSCAN).

4.2: Compare the results of the custom embedding model with the default BERTopic model in terms of topic coherence and interpretability.

4.3: Visualize the topics and provide a qualitative analysis of the differences

**

Usefull Link :<https://www.pinecone.io/learn/bertopic/>

```
import pandas as pd
import random
from sklearn.datasets import fetch_20newsgroups

# Load dataset and sample 2000 rows
data = fetch_20newsgroups(subset='all', remove=('headers', 'footers', 'quotes'))
sampled_data = random.sample(data.data, 2000)

# Convert to DataFrame
dataframe_3 = pd.DataFrame(sampled_data, columns=['text'])
print(dataframe_3.head())
```

```

text
0  I am looking for a WIN31 driver (or set) for m...
1  \n: |>You might visit some congregations of Ch...
2  I would like to know the regular/international...
3  Version 1.3 of Xew widgets is available at\n\n...
4  \n\nYou are absolutely daft. No flame require...
```

```
!pip install bertopic
```

```

Requirement already satisfied: bertopic in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: hdbscan>=0.8.29 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: pandas>=1.1.5 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: scikit-learn>=1.0 in /usr/local/lib/python3.11/dist
Requirement already satisfied: sentence-transformers>=0.4.1 in /usr/local/lib/pyth
Requirement already satisfied: tqdm>=4.41.1 in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: umap-learn>=0.5.0 in /usr/local/lib/python3.11/dist
Requirement already satisfied: scipy>=1.0 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-package
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/d
Requirement already satisfied: transformers<5.0.0,>=4.41.0 in /usr/local/lib/pytho
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.1
Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.
Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.4.127 in /usr/local/lib/
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.4.127 in /usr/local/li
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.4.127 in /usr/local/lib/
Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in /usr/local/lib/pytho
Requirement already satisfied: nvidia-cublas-cu12==12.4.5.8 in /usr/local/lib/pyth
```

```
Requirement already satisfied: nvidia-cufft-cu12==11.2.1.3 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: nvidia-curand-cu12==10.3.5.147 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: nvidia-cusolver-cu12==11.6.1.9 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: nvidia-cusparse-cu12==12.3.1.170 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: nvidia-nvjitlink-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages
```

```
!pip install openai==0.27.8
```



```
Collecting openai==0.27.8
```

```
Using cached openai-0.27.8-py3-none-any.whl.metadata (13 kB)
```

```
Requirement already satisfied: requests>=2.20 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from openai==0.27.8)
Requirement already satisfied: aiohttp in /usr/local/lib/python3.11/dist-packages (from openai==0.27.8)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-packages
Using cached openai-0.27.8-py3-none-any.whl (73 kB)
```

```
Installing collected packages: openai
```

```
Attempting uninstall: openai
```

```
Found existing installation: openai 1.71.0
```

```
Uninstalling openai-1.71.0:
```

```
Successfully uninstalled openai-1.71.0
```

```
Successfully installed openai-0.27.8
```

```
!pip install 'numpy>=1.24'
```



```
Requirement already satisfied: numpy>=1.24 in /usr/local/lib/python3.11/dist-packages
```

```
!pip install --upgrade numpy --quiet
!pip uninstall -y bertopic
!pip install bertopic[all] --quiet
```

ERROR: pip's dependency resolver does not currently take into account all the package dependencies that were installed when a package was added to your environment. This is normal for packages built by pip, or untrusted sources like wheel. It can be enabled using pip's option: 'root-user-action: yes' (see PEP 668).

tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.2.4 which is incompatible.

gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.2.4 which is incompatible.

google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3 which is incompatible.

tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow 2.19.0 which is incompatible.

numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.2.4 which is incompatible.

tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which is incompatible.

Found existing installation: bertopic 0.17.0
Uninstalling bertopic-0.17.0:
Successfully uninstalled bertopic-0.17.0

WARNING: bertopic 0.17.0 does not provide the extra 'all'

ERROR: pip's dependency resolver does not currently take into account all the package dependencies that were installed when a package was added to your environment. This is normal for packages built by pip, or untrusted sources like wheel. It can be enabled using pip's option: 'root-user-action: yes' (see PEP 668).

gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.0.2 which is incompatible.

google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.2.3 which is incompatible.

tensorflow-text 2.18.1 requires tensorflow<2.19,>=2.18.0, but you have tensorflow 2.19.0 which is incompatible.

tf-keras 2.18.0 requires tensorflow<2.19,>=2.18, but you have tensorflow 2.19.0 which is incompatible.

```
!pip install --upgrade jax jaxlib
```

Requirement already satisfied: jax in /usr/local/lib/python3.11/dist-packages (0.5.3)

Requirement already satisfied: jaxlib in /usr/local/lib/python3.11/dist-packages (0.5.3)

Requirement already satisfied: ml_dtypes>=0.4.0 in /usr/local/lib/python3.11/dist-packages (0.5.3)

Requirement already satisfied: numpy>=1.25 in /usr/local/lib/python3.11/dist-packages (1.26.0)

Requirement already satisfied: opt_einsum in /usr/local/lib/python3.11/dist-packages (3.3.0)

Requirement already satisfied: scipy>=1.11.1 in /usr/local/lib/python3.11/dist-packages (1.11.2)

```
!pip install bertopic[all]
!pip install --upgrade sentence-transformers
!pip install --upgrade jax jaxlib
```

Requirement already satisfied: bertopic[all] in /usr/local/lib/python3.11/dist-packages (0.17.0)

WARNING: bertopic 0.17.0 does not provide the extra 'all'

Requirement already satisfied: hdbscan>=0.8.29 in /usr/local/lib/python3.11/dist-packages (0.8.29)

Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.11/dist-packages (1.26.0)

Requirement already satisfied: pandas>=1.1.5 in /usr/local/lib/python3.11/dist-packages (2.2.3)

Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.11/dist-packages (5.22.0)

Requirement already satisfied: scikit-learn>=1.0 in /usr/local/lib/python3.11/dist-packages (1.5.2)

Requirement already satisfied: sentence-transformers>=0.4.1 in /usr/local/lib/python3.11/dist-packages (3.0.1)

Requirement already satisfied: tqdm>=4.41.1 in /usr/local/lib/python3.11/dist-packages (4.67.1)

Requirement already satisfied: umap-learn>=0.5.0 in /usr/local/lib/python3.11/dist-packages (0.5.7)

Requirement already satisfied: scipy>=1.0 in /usr/local/lib/python3.11/dist-packages (1.11.2)

Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.11/dist-packages (1.4.2)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (2024.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (2024.2)

Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.11/dist-packages (9.0.0)

Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (24.1)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/dist-packages (3.5.0)

Requirement already satisfied: transformers<5.0.0,>=4.41.0 in /usr/local/lib/python3.11/dist-packages (4.41.0)

Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.11/dist-packages (2.5.1)

Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.11/dist-packages (0.24.7)

```

Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.
Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.4.127 in /usr/local/lib/
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.4.127 in /usr/local/li
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.4.127 in /usr/local/lib/
Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in /usr/local/lib/pytho
Requirement already satisfied: nvidia-cublas-cu12==12.4.5.8 in /usr/local/lib/pyth
Requirement already satisfied: nvidia-cufft-cu12==11.2.1.3 in /usr/local/lib/pytho
Requirement already satisfied: nvidia-curand-cu12==10.3.5.147 in /usr/local/lib/py
Requirement already satisfied: nvidia-cusolver-cu12==11.6.1.9 in /usr/local/lib/py
Requirement already satisfied: nvidia-cusparse-cu12==12.3.1.170 in /usr/local/lib/
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/pyt
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python
Requirement already satisfied: nvidia-nvjitlink-cu12==12.4.127 in /usr/local/lib/p
Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dis
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist
Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.11
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dis
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-p
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-pack
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dis
Requirement already satisfied: certifi<2017.4.17 in /usr/local/lib/python3.11/dis

```

```

!pip install --upgrade jax jaxlib
!pip install --upgrade tensorflow

```



```

Requirement already satisfied: jax in /usr/local/lib/python3.11/dist-packages (0.5.3)
Requirement already satisfied: jaxlib in /usr/local/lib/python3.11/dist-packages (0.5
Requirement already satisfied: ml_dtypes>=0.4.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: numpy>=1.25 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: opt_einsum in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: scipy>=1.11.1 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: tensorflow in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.11/dist
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/
Requirement already satisfied: google-pasta>=0.1.1 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: setuptools in /usr/local/lib/python3.11/dist-packages

```